Identification of protein targets of the humoral immune response in Alzheimer's disease combining immunoprecipitation with phage display and LC-MS/MS

alzheimer's association AAIC 17

P3-245

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INTRODUCTION and OBJECTIVES

Alzheimer's disease (AD) is a progressive and chronic neurodegenerative disorder that affects wide areas of the cerebral cortex and hippocampus. It is the most common form of dementia worldwide, with a 10-30% prevalence in the ageing population (>65 years of age), a mean incidence of 1-3% and a high socioeconomic impact. The clinical phases are preceded by preclinical and prodromal stages, which are extended over 2-3 decades[1] before the first symptoms. AD patients' brains are characterized by two main hallmarks: the formation of intracellular neurofibrillary tangles of hyperphosphorylated Tau and the presence of extracellular amyloid plaques. The specific detection of AD, for which definitive diagnosis requires *post mortem* verification, includes molecular PET imaging, measurement of Aβ and tau in the cerebrospinal fluid and behavior history assessments. However, these methods are not enough for early detection. Therefore, new diagnostic biomarkers and therapeutic targets of intervention are necessary.

Due to the drastic changes in AD brains caused by neurodegeneration, the blood brain barrier is altered and the brain enters into a permanent inflammatory state. These may lead to the generation of AD-specific autoantibodies.

Our objective is the identification of their target proteins, which might become a new source of AD biomarkers and a new approach to study AD preclinical and clinical phases at the cellular and molecular levels.

Identifying AD-specific autoantibodies by IP and LC-MS/MS

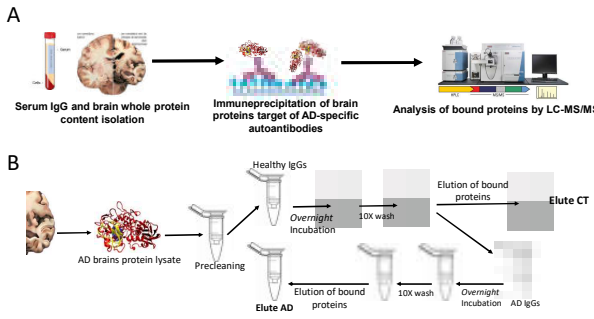


Figure 1. Work-flow to identify AD autoantibodies by LC-MS/MS. The whole protein content of AD patients' brain was incubated with protein-G bound IgGs from healthy individuals' and AD patients' sera and the especifically bound proteins were analyzed by LC-MS/MS (**A**). Firstly, the protein lysate from AD patients' brains was preincubated with protein G and with the total IgG from healthy patients. Finally, the precleared lysate was incubated with IgGs from AD patients. Proteins bound to healthy and AD-specific IgGs were finally analyzed (**B**).

Identifying AD-specific autoantibodies by T7 phage microarrays

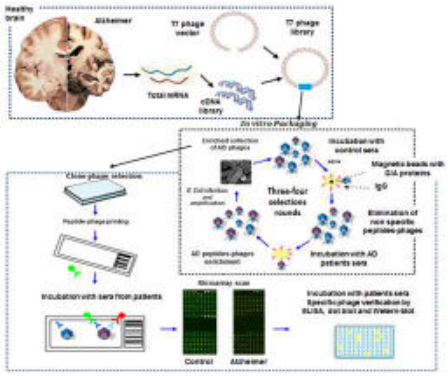


Figure 2. Scheme of the work-flow to identify AD autoantibodies using phage display and protein microarrays. Construction of the two T7 phage libraries displaying the cDNA repertoire of AD patients and healthy individuals' brain by phage display. Both libraries were biopanned using serum from healthy individuals and subsequently with AD patients' sera. The control library let us to survey target autoantibodies for wild type specific proteins in AD in contrast to other dementias, whereas AD library let us to identify target of autoantibodies proteins alterations (alternative splicing, chimera or frameshift) associated with the ongoing of the disease.

The identification of specific phages displaying proteins or peptides recognized by IgGs from AD patients is then achieved after printing nitrocellulose microarray with 768 phages, which are then probed with sera from controls and AD patients at different Braak stages.

LC-MS/MS data analysis

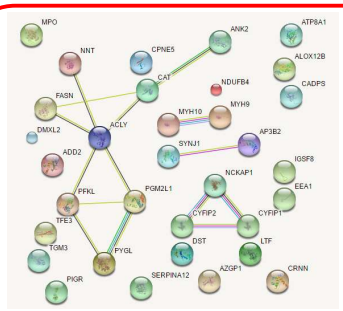


Figure 3. Protein interaction map of the 40 proteins recognized by AD-specific autoantibodies and identified by LC-MS/MS. STRING 10.5 analysis of known and predicted protein-protein interactions showed two main altered axis: one related to glucose, fatty-acid and ROS metabolism, and another one related to the WASP-WAVE signaling pathway. Gene Ontology analysis of molecular function showed that the identified proteins were mainly involved in metabolic regulation, synaptic vesicle formation, endocytosis and actin remodeling. Moreover, previous studies have shown the altered expression of six of these proteins and described the functional relationship between nearly half of them and pathways related to AD devolpment.

Isolation and insert sequencing of the immunoreactive phages

Phage	Displayed peptide	HIT Blast p hit (*Homo sapiens*)
8-3AM	NSPKQFSAG	NP_003469.2
10-3AM	NSSCINKIDFHNKMVYPQIKIKNN	NP_113676.2
4-4AM	NSRISYTSILNHFYCHNNSSISYTSIL NHFYCHN	NP_001005205.2
3-4AM	NSSLRLPGSRHSPASASRVVGTTG THHHARLMFCIFSRDGVSLC	NP_001164252.1
55-4AS	NSSPPCSEPSAFWALRL	NP_006346.1
9-4AS	NSERKKRGARR	NP_036517.1
82-4cs	NSYWRETLQM	NP_001306055.1
80-3as	NSKEVVWKEKFTDTR	NP_001292721.1
5-4cs	NSSRSLIVLGQ	NP_005838.3
pep2	NSSLKTHLVPESCPQLLNITALKL	NP_003469.2

Table 1. Peptides displayed on phages and their Blast p hit with the smallest p-Value. The cDNA inserts of the most immunoreactive phages were PCR-amplified and sequenced. The identified encoded peptides were then compared to human proteins on NCBI data base using Blat p. Most phages showed high sequence identity with different proteins, including olfactory receptors, transcription factors, solute carriers and proteins involved in proteasome regulation. Moreover, two phages were displaying different peptides form the same E3-Ubiquitin Ligase.

DATA VALIDATION

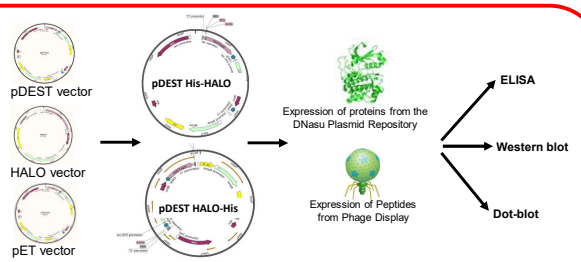


Figure 4. Firstly, two Destination (pDEST) vectors were designed and constructed for the expression of N- and C-terminal His and Halo-tagged proteins and phage-derived peptides. Then, the immunoreactive peptides and proteins were cloned, expressed and purified using a Ni-NTA resin.

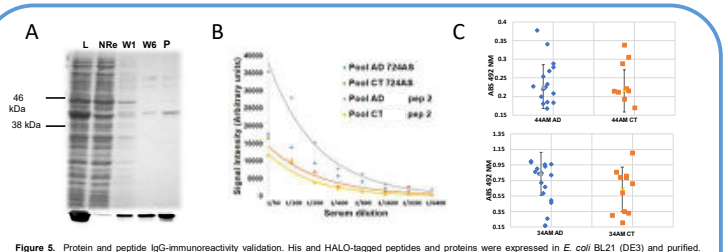


Figure 5. Protein and peptide IgG-immunoreactivity validation. His and HALO-tagged peptides and proteins were expressed in *E. coli* BL21 (DE3) and purified. Protein purity and integrity were analyzed by Coomassie Blue staining after SDS-PAGE and WB with an anti-6xHis mAb: P, purified protein (A). Analysis of the specific humoral response to indicated peptides fused to Halo-tag. Pool of Alzheimers' Disease (AD) patients' or controls' (CT) sera were incubated with HaloTag magnetic beads in solution at the indicated dilutions and bound human IgG detected with a secondary anti human IgG Biotin-labeled Ab followed by streptavidin-peroxidase incubation. The signal was developed by luminiscence. 724AS, negative control (B). GST-tagged peptides 44AM and 34AM were analyzed by ELISA at 1:100 dilution of individual sera form AD and CT individuals followed by anti-Human IgG HRP-labeld antibody (C).

Conclusions

→ A total of 50 peptides and proteins have been identified as potential targets of serum autoantibodies biomarkers specific for Alzheimer's Disease.
→ The three phage displayed peptides tested by ELISA so far show more IgG-recognition by AD patients' sera than healthy individuals.
→ Serum IgG-mediated Immunoprecipitation and T7 Phage Display in combination with LC-MS/MS and protein microarrays are useful techniques to identifiy potentital candidates for disease-specific blood biomarkers.

Future prospects

To confirm the usefulness of these proteins as specific AD biomarkers, we will use a higher cohort of human sera samples, including healthy individuals and patients with other dementias.

We will also analyze if these proteins show altered expression in different brain regions, especially those early affected in the course of the disease, and if the play an important functional role in AD pathogenesis.

Pablo San Segundo has a FPU. Rodrigo Barderas acknowledges the support of Ramón y Cajal programme.